US012411438B2

(12) United States Patent
Adams et al.

(10) Patent No.: US 12,411,438 B2
(45) Date of Patent: Sep. 9, 2025

(54) **TONER CONTAINER HANDLE**

(71) Applicant: **LEXMARK INTERNATIONAL, INC.**, Lexington, KY (US)

(72) Inventors: **Emily Stefano Adams**, Lexington, KY (US); **Brian Lester Boettcher**, Versailles, KY (US); **Patrick Lawrence Ransom**, Lexington, KY (US)

(73) Assignee: **Lexmark International, Inc.**, Lexington, KY (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/762,008**

(22) Filed: **Jul. 2, 2024**

(65) **Prior Publication Data**

US 2024/0353774 A1 Oct. 24, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/961,093, filed on Oct. 6, 2022, now Pat. No. 12,055,872.

(60) Provisional application No. 63/406,831, filed on Sep. 15, 2022.

(51) **Int. Cl.**
G03G 15/08 (2006.01)
G03G 21/18 (2006.01)

(52) **U.S. Cl.**
CPC ***G03G 15/0875*** (2013.01); ***G03G 21/1842*** (2013.01); ***G03G 21/1846*** (2013.01); *G03G 2215/068* (2013.01); *G03G 2221/1846* (2013.01)

(58) **Field of Classification Search**
CPC G03G 15/0875; G03G 21/1846; G03G 2221/1846; G03G 21/1853; G03G 21/1839; G03G 21/1842; G03G 2215/068
USPC D18/43
See application file for complete search history.

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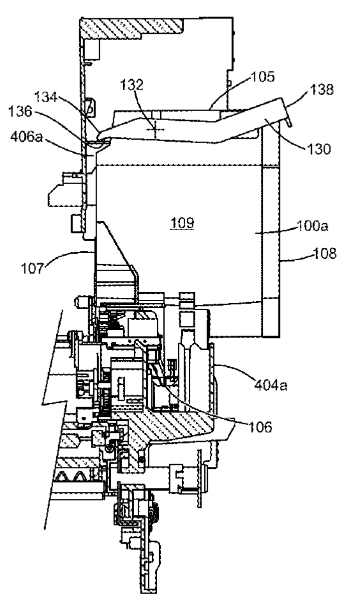
Primary Examiner — Stephanie E Bloss
Assistant Examiner — Laura Roth
(74) *Attorney, Agent, or Firm* — Justin M. Tromp

(57) **ABSTRACT**

A toner container according to one example embodiment includes a pivotable handle. The handle includes a front segment extending along a front of a body of the toner container when the handle is in a first position for contacting a corresponding pry surface in an image forming device when the handle moves from the first position to a second position. The handle includes a rear segment extending along a rear of the body when the handle is in the first position permitting a user to actuate the rear segment to move the handle from the first position to the second position. The handle includes a first side segment extending along the first side of the body and a second side segment extending along the second side of the body. The first and second side segments connect the front segment and the rear segment.

11 Claims, 14 Drawing Sheets



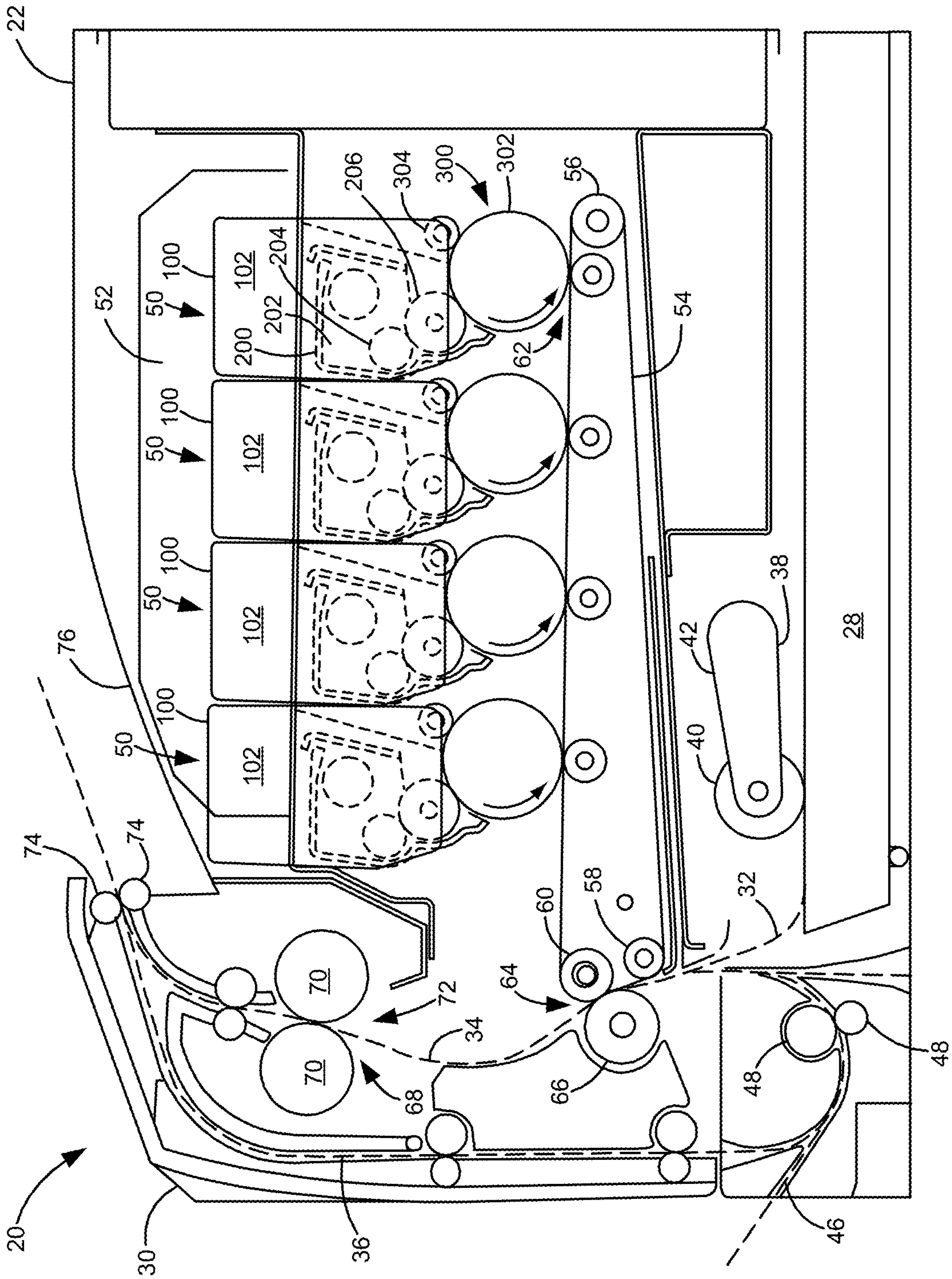

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Figure 1

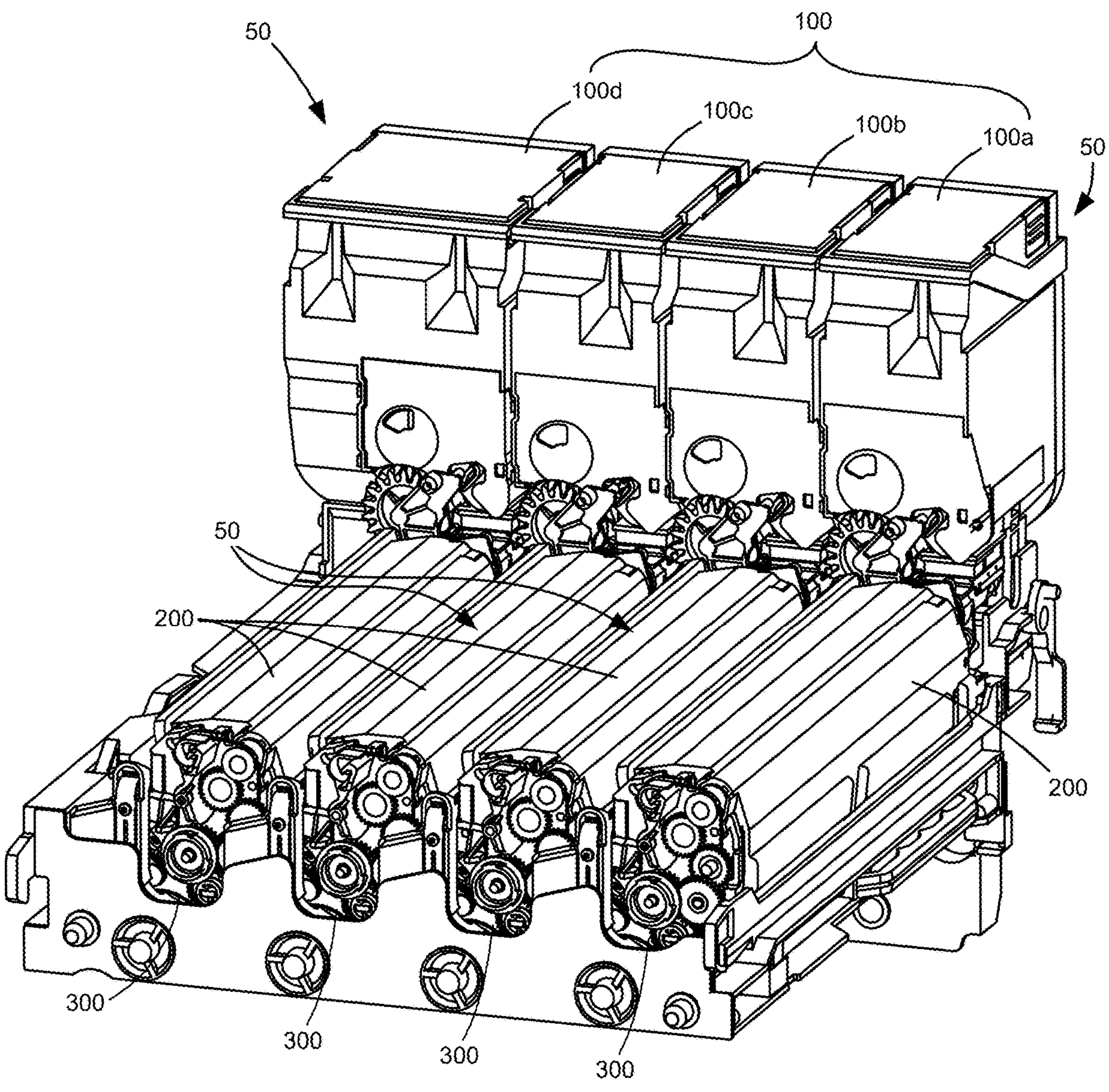
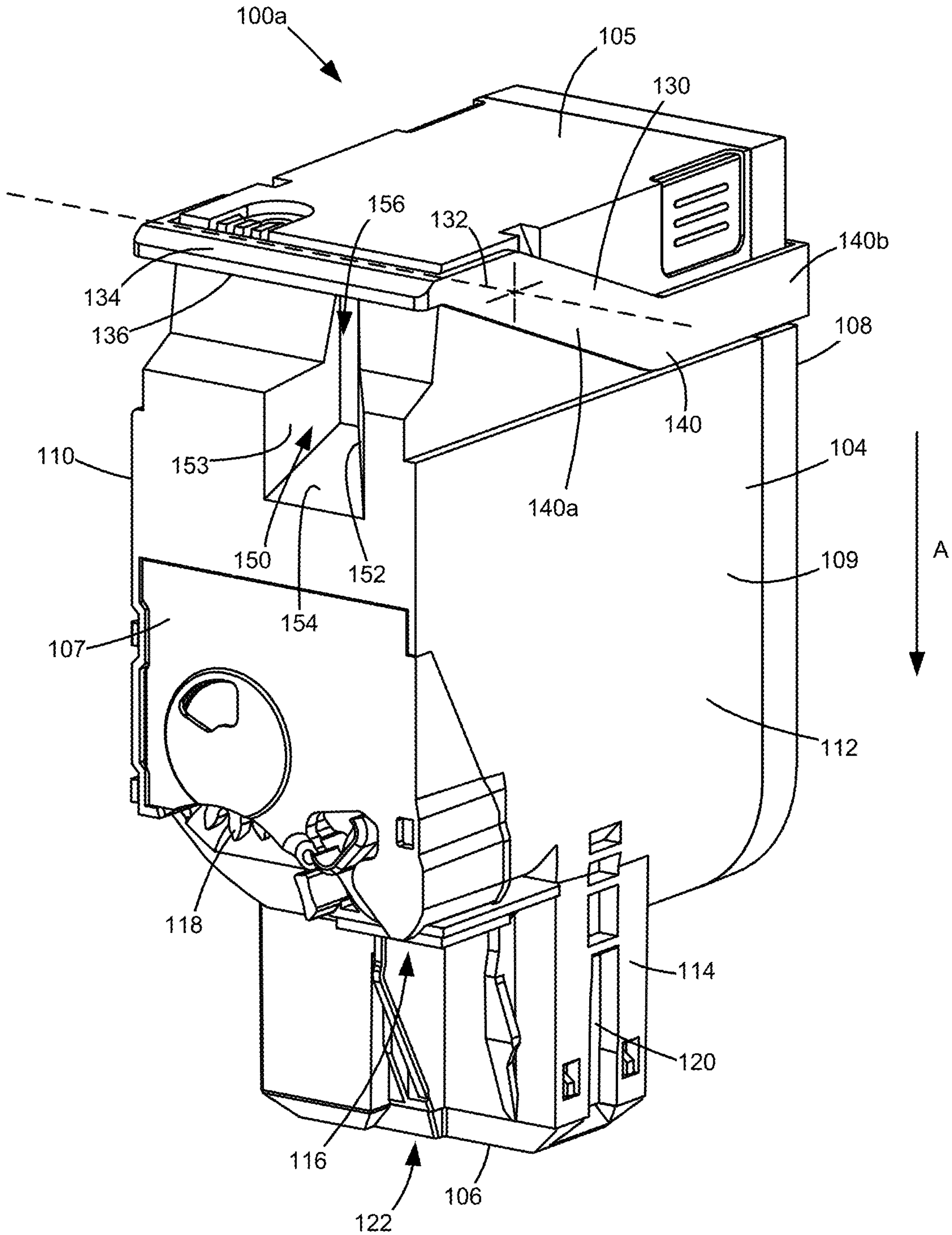


Figure 2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140b
134
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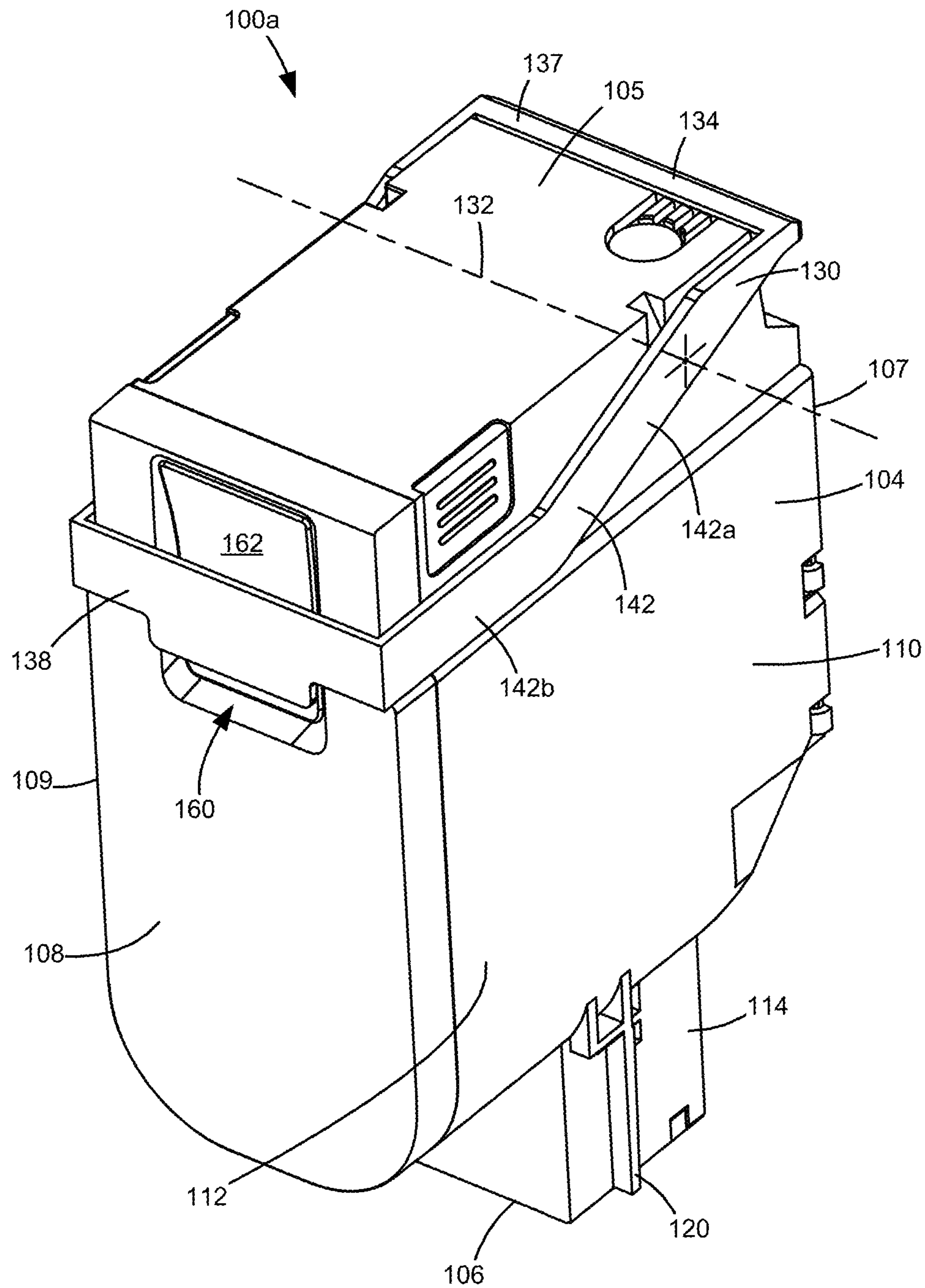
120
116
106
122

Figure 3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106

Figure 4

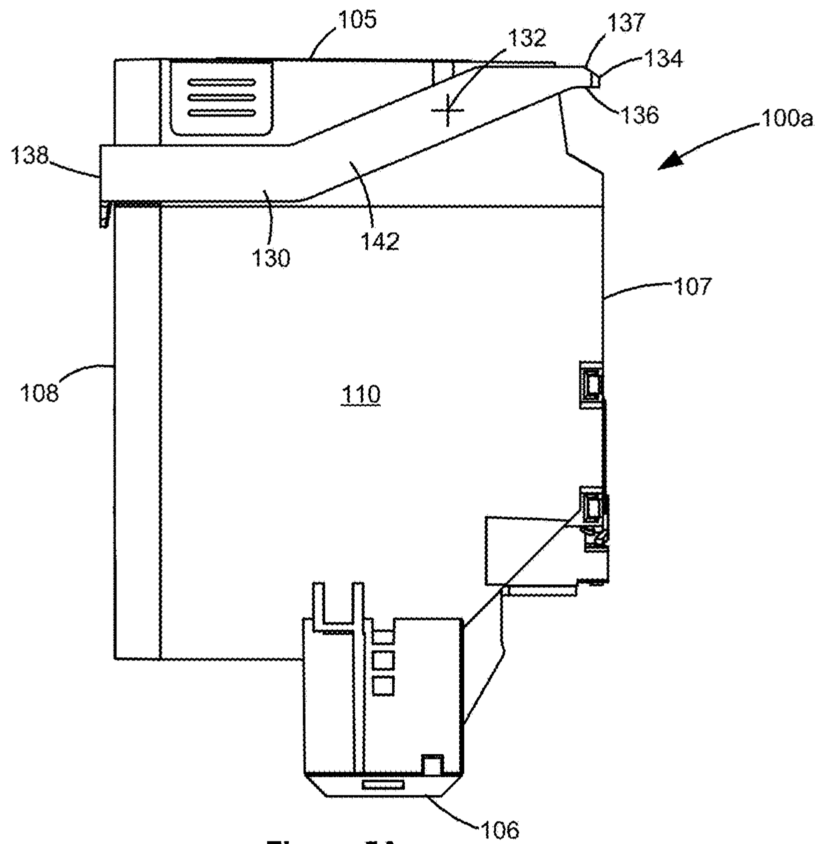


Figure 5A

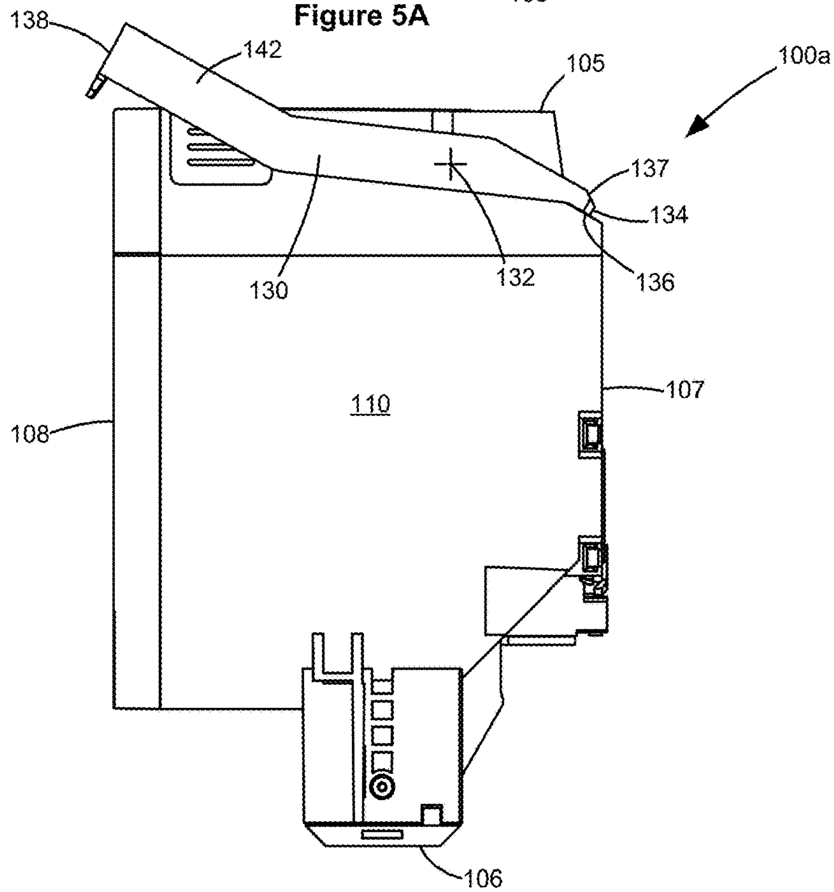
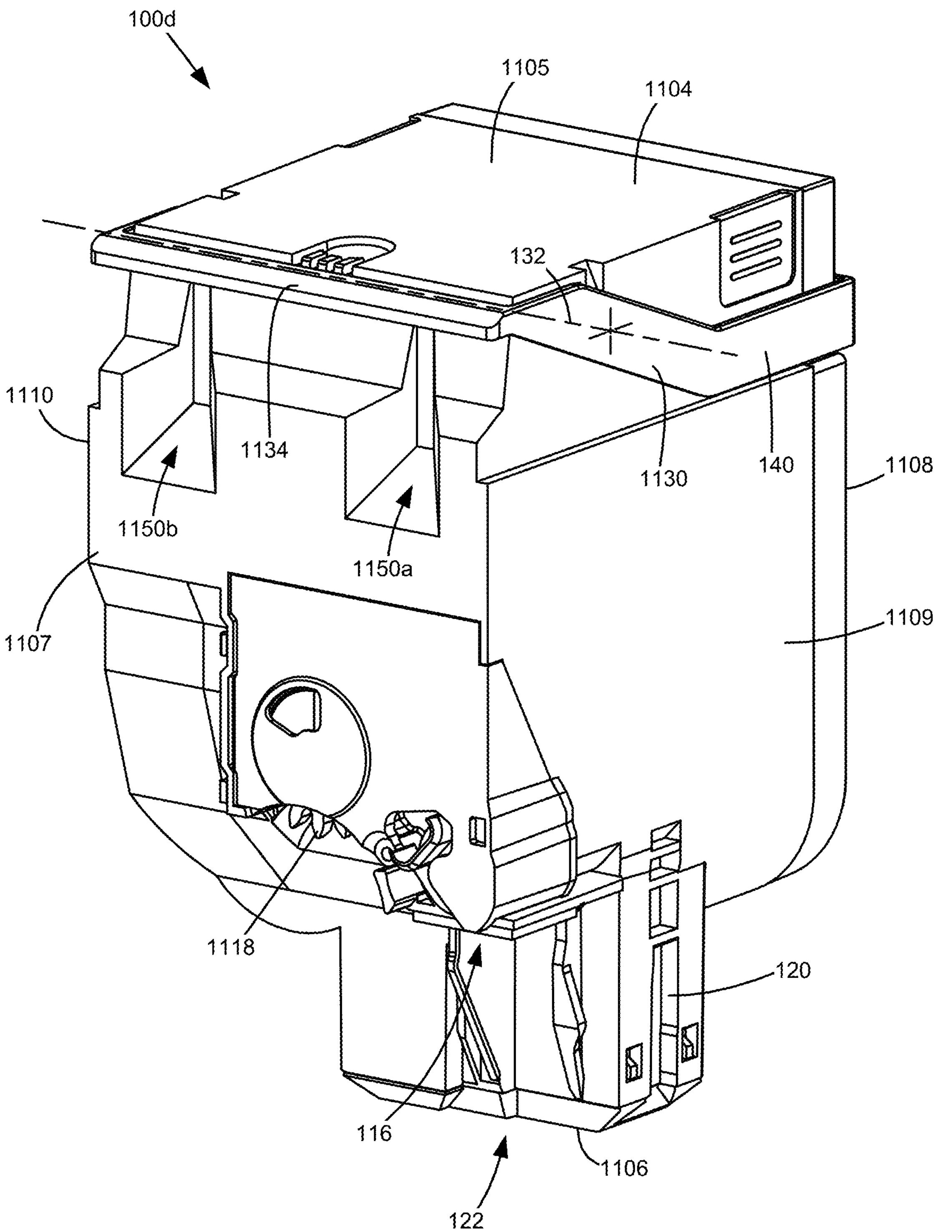


Figure 5B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116
1106
122

Figure 6

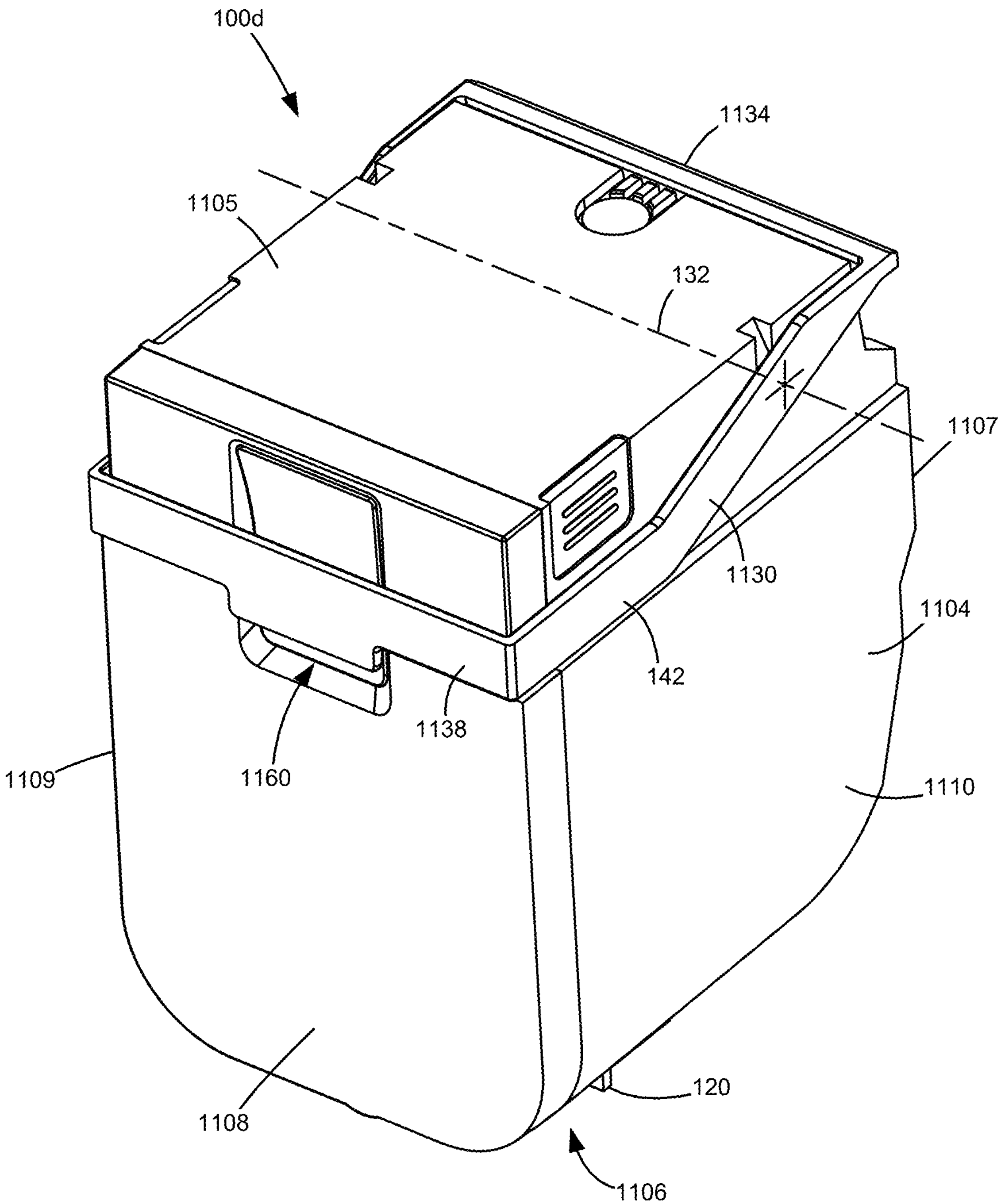
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Figure 7

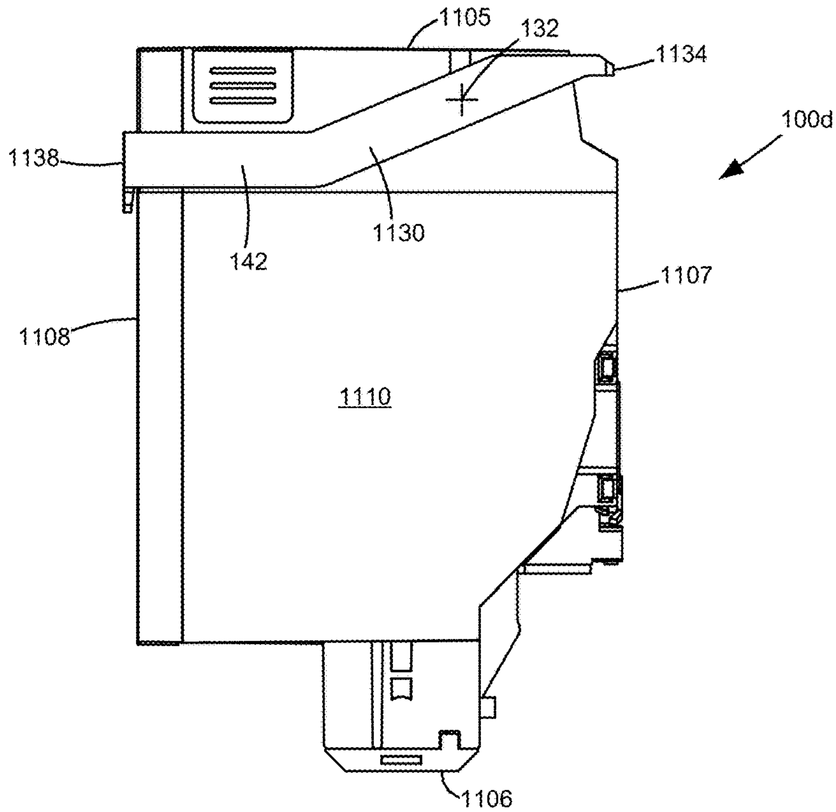

Figure 8A

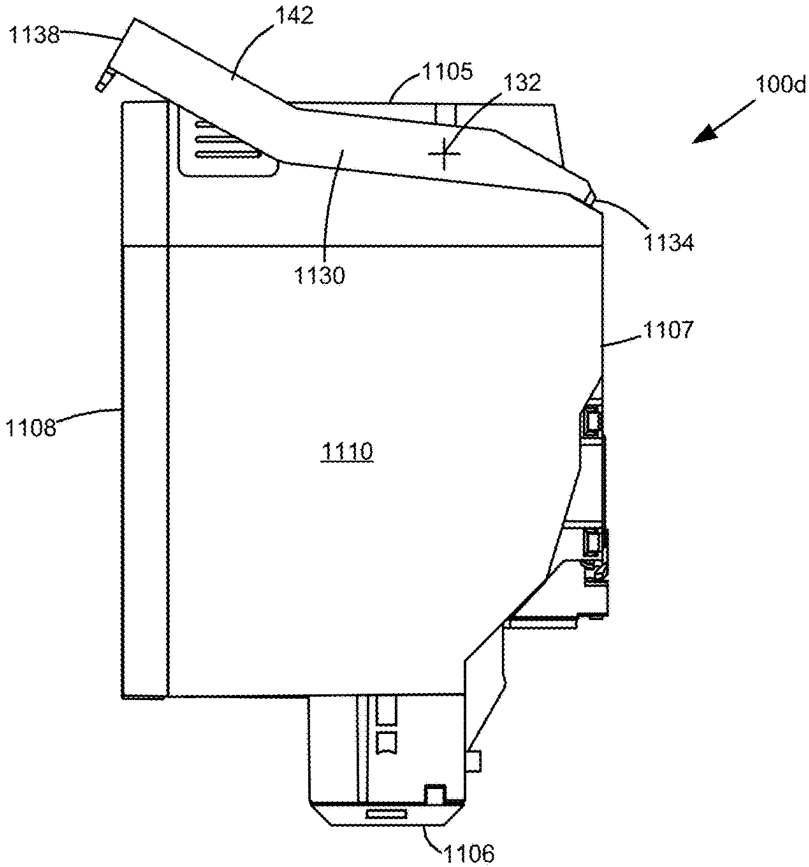

Figure 8B

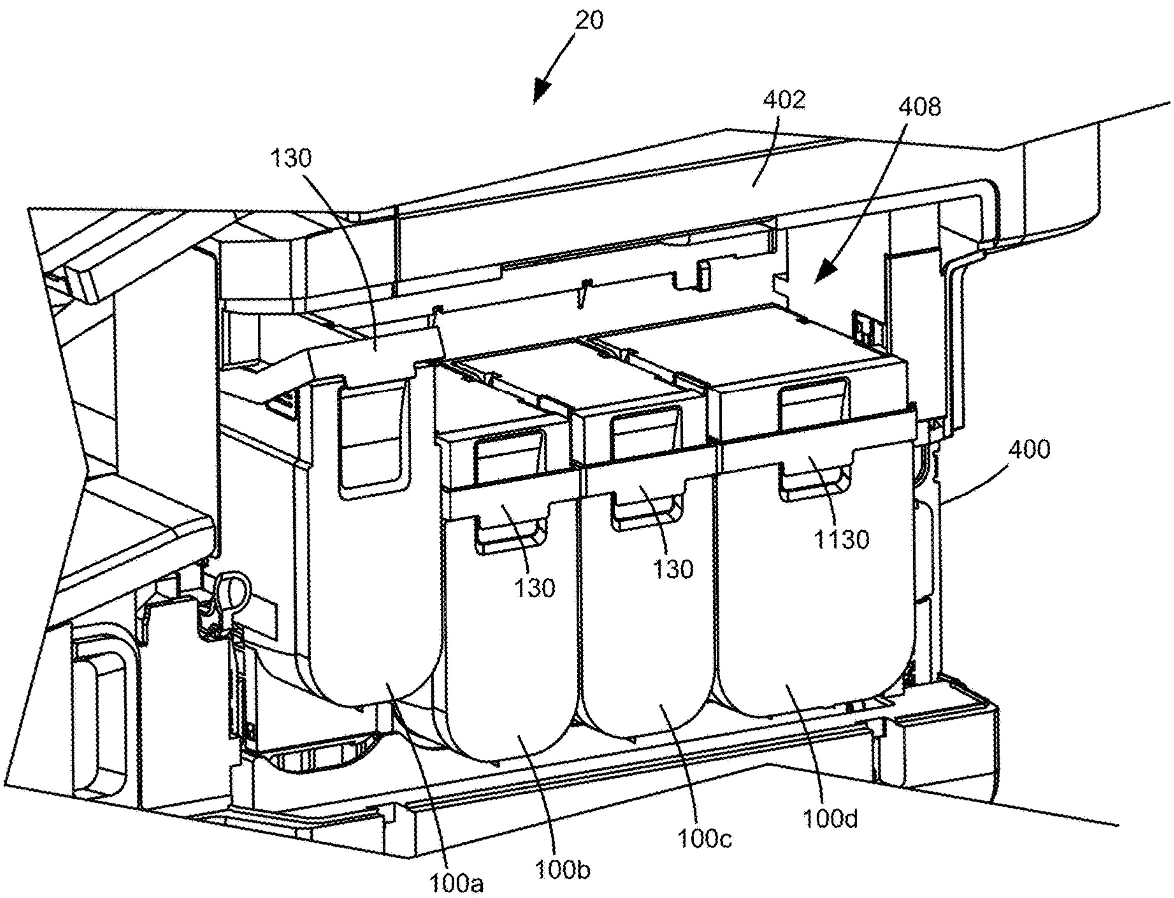

Figure 9

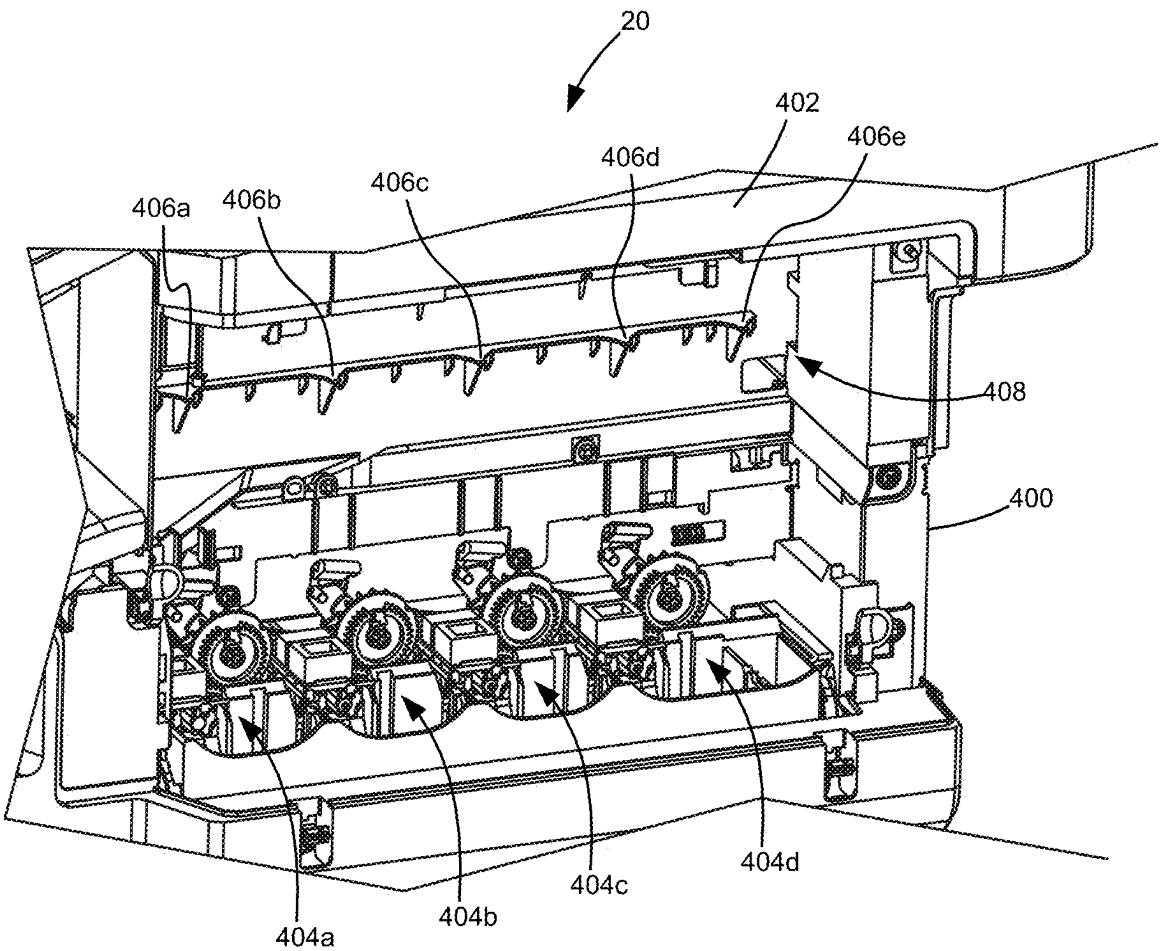

Figure 10

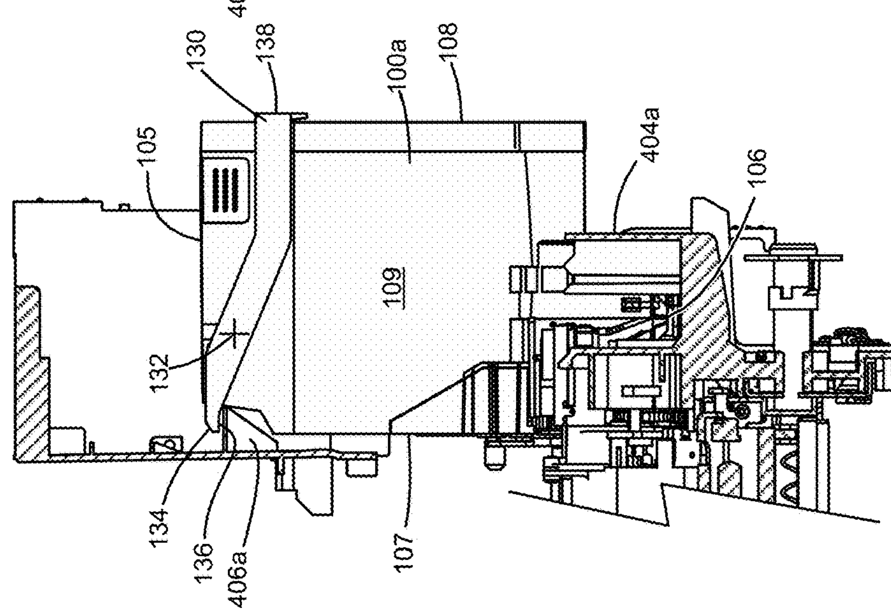

Figure 11A

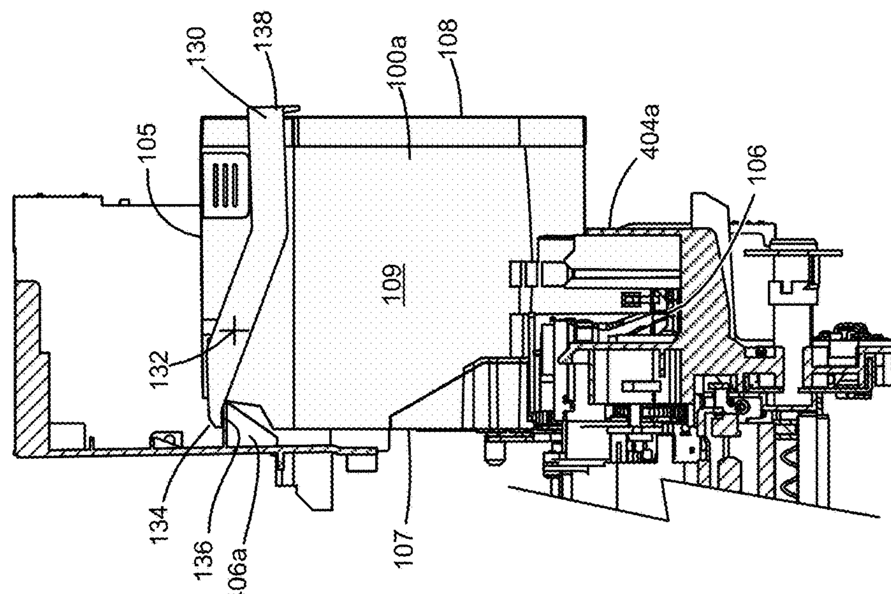

Figure 11B

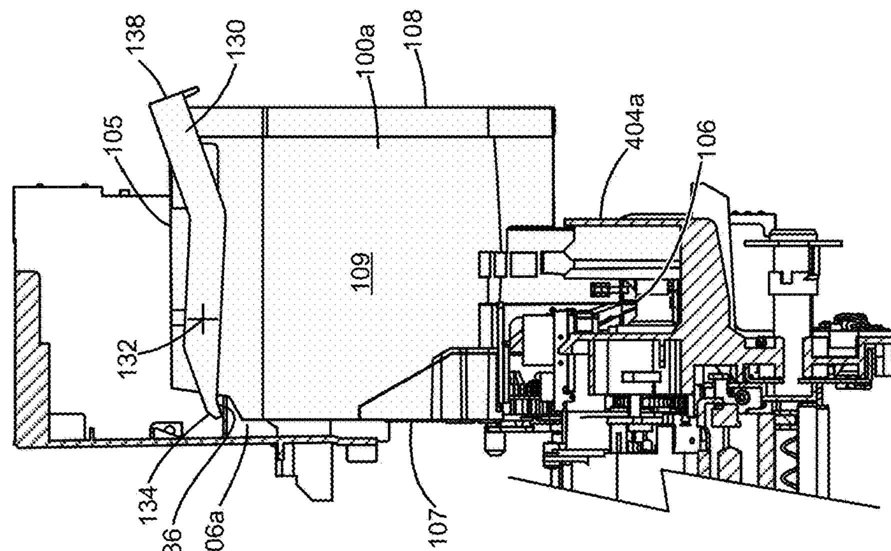

Figure 11C

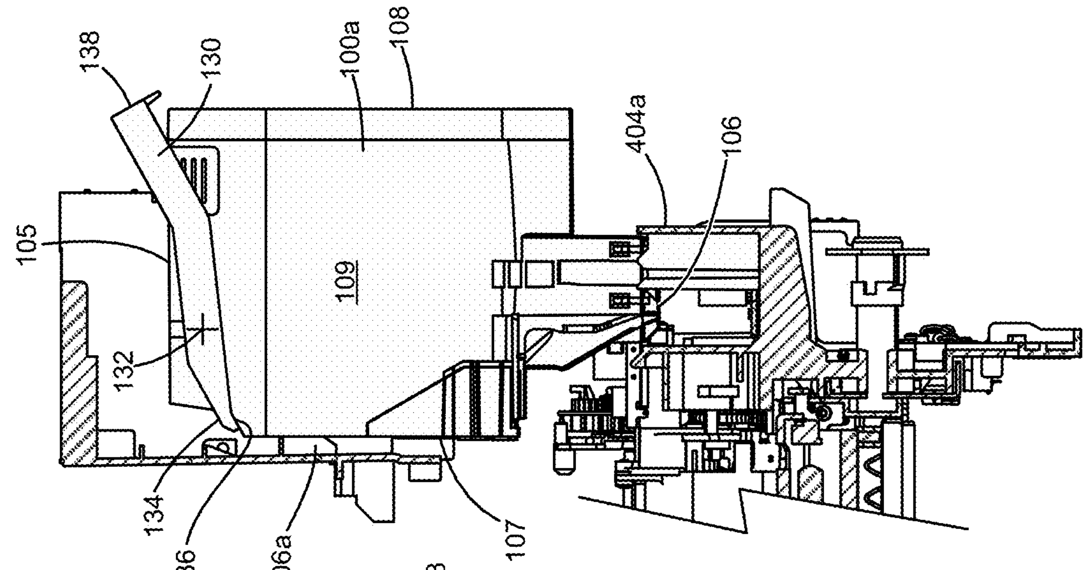

Figure 11F

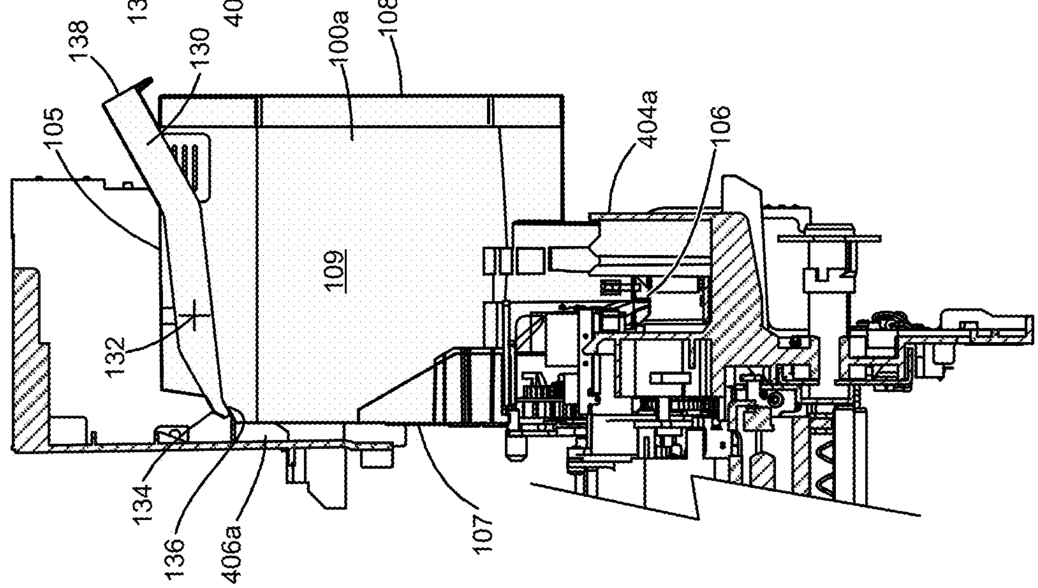

Figure 11E

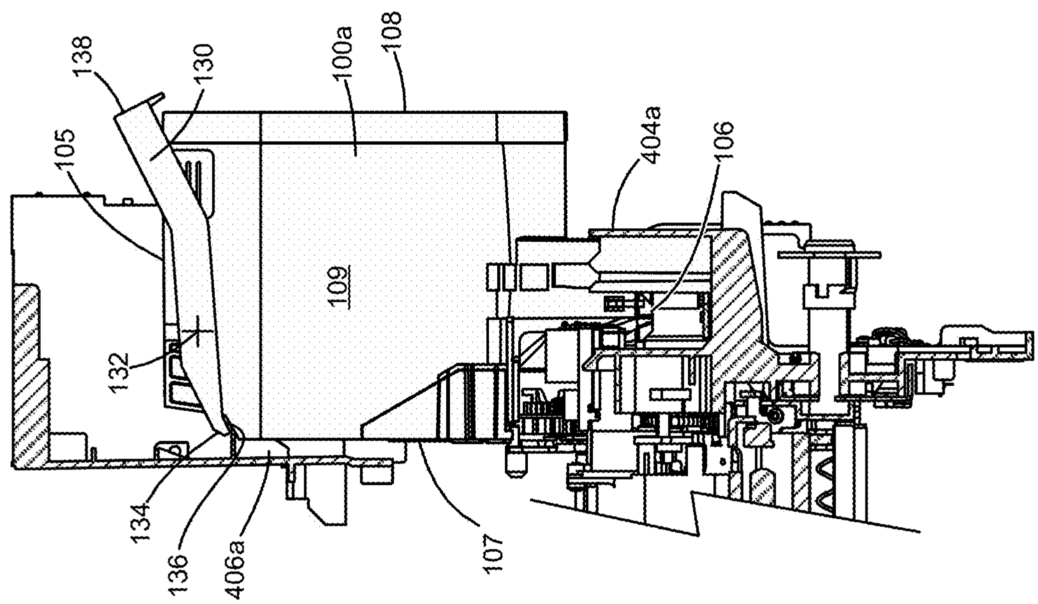

Figure 11D

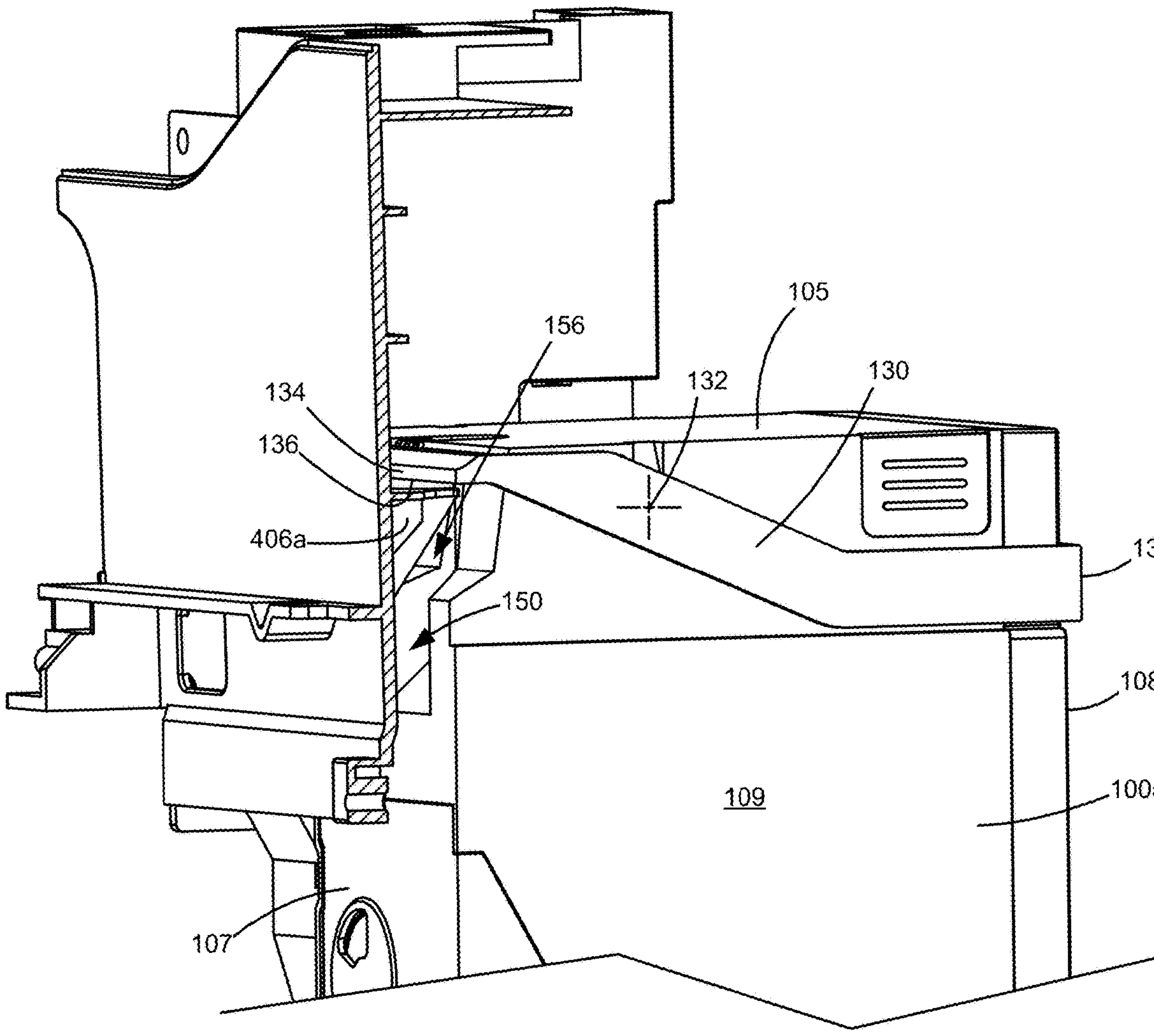

Figure 12

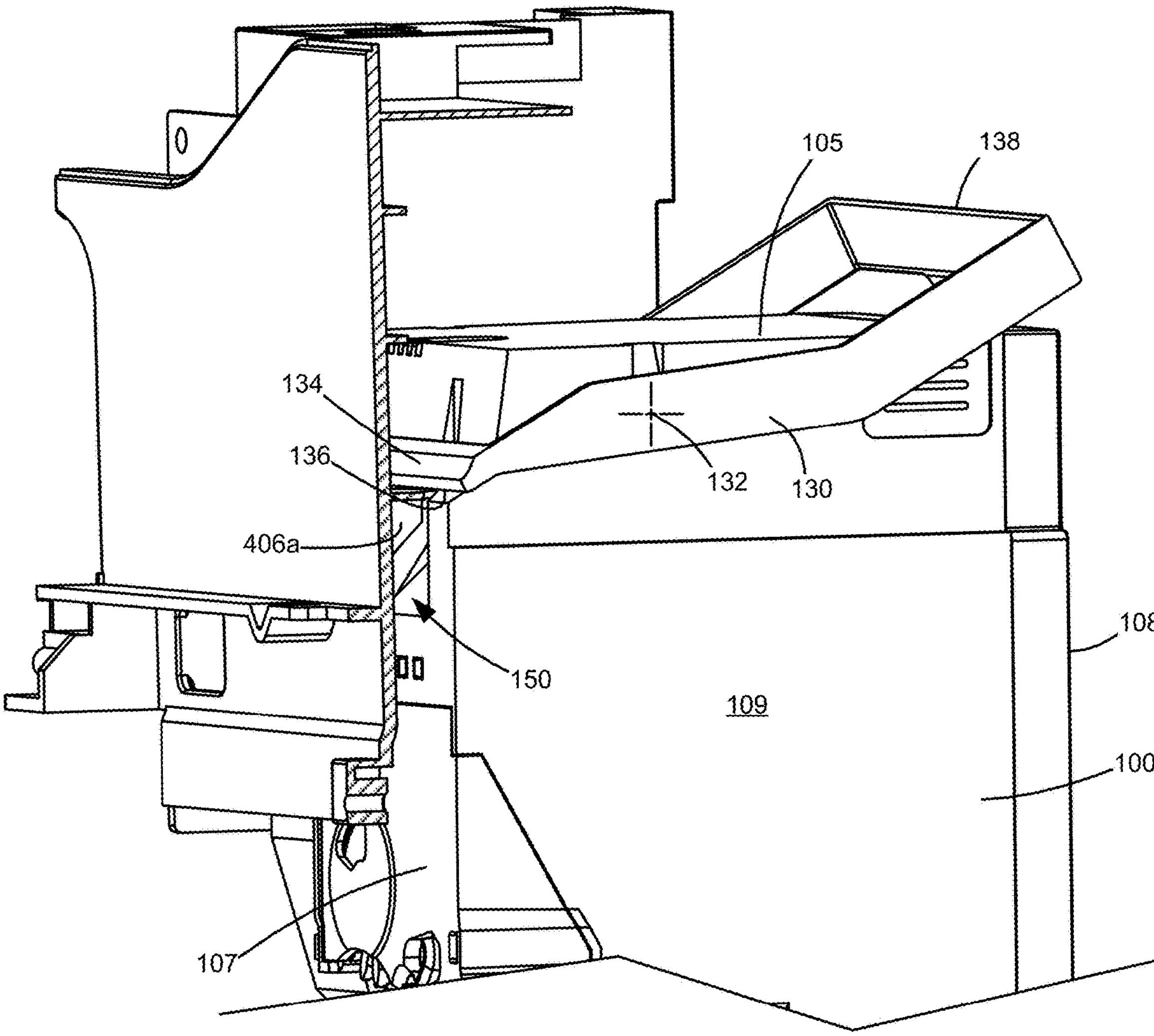

Figure 13

TONER CONTAINER HANDLE

CROSS REFERENCES TO RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 17/961,093, filed Oct. 6, 2022, entitled "Toner Container Handle," which claims priority to U.S. Provisional Patent Application Ser. No. 63/406,831, filed Sep. 15, 2022, entitled "Toner Container Handle," the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

1. Field of the Disclosure

The present disclosure relates generally to image forming devices and more particularly to a toner container handle.

2. Description of the Related Art

During the electrophotographic printing process, an electrically charged rotating photoconductive drum is selectively exposed to a laser beam. The areas of the photoconductive drum exposed to the laser beam are discharged creating an electrostatic latent image of a page to be printed on the photoconductive drum. Toner particles are then electrostatically picked up by the latent image on the photoconductive drum creating a toned image on the drum. The toned image is transferred to the print media (e.g., paper) either directly by the photoconductive drum or indirectly by an intermediate transfer member. The toner is then fused to the media using heat and pressure to complete the print.

The image forming device's toner supply is typically stored in one or more replaceable units that have a shorter lifespan than the image forming device. The replaceable units require periodic replacement by the user in order to continue printing. The user experience associated with replacing the replaceable unit is an important design consideration. Preferably, the process of removing an existing replaceable unit and installing a replacement is intuitive, efficient and relatively simple. From the manufacturer's perspective, it is desired to reduce the cost and complexity of the replaceable unit and overall system design. It is also desired to minimize the overall size of the replaceable unit and the image forming device, consistent with consumer preferences for smaller devices and components.

SUMMARY

A toner container for use in an image forming device according to one example embodiment includes a body having a top, a bottom, a front and a rear positioned between a first side and a second side of the body. The body has a reservoir for holding toner. A downward facing outlet port is positioned on the front of the body for exiting toner from the toner container. A handle is pivotable relative to the body about a pivot axis between a first position and a second position. The handle includes a front segment extending along the front of the body when the handle is in the first position for contacting a corresponding pry surface in the image forming device when the handle moves from the first position to the second position for lifting the toner container from a corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device. The handle includes a rear segment

extending along the rear of the body when the handle is in the first position permitting a user to actuate the rear segment to move the handle from the first position to the second position for removing the toner container from the corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device. The handle includes a first side segment extending along the first side of the body and a second side segment extending along the second side of the body. The first and second side segments connect the front segment and the rear segment.

Embodiments include those wherein the front segment, the rear segment and the first and second side segments of the handle are positioned no higher than the top of the body when the handle is in the first position.

In some embodiments, the front segment extends horizontally along the front of the body when the handle is in the first position.

Embodiments include those wherein a bottom contact surface of the front segment positioned to contact the corresponding pry surface in the image forming device when the handle moves from the first position to the second position is positioned higher than the pivot axis when the handle is in the first position and lower than the pivot axis when the handle is in the second position.

In some embodiments, the rear segment extends horizontally along the rear of the body when the handle is in the first position.

Embodiments include those wherein the rear segment is positioned lower than the pivot axis when the handle is in the first position and higher than the pivot axis when the handle is in the second position.

In some embodiments, the first and second side segments each include a front portion that is proximate to the front of the body and that extends downward and rearward when the handle is in the first position. In some embodiments, the pivot axis extends through the front portions of the first and second side segments.

Embodiments include those wherein the first and second side segments each include a rear portion that extends horizontally when the handle is in the first position.

Some embodiments include a notch in the front of the body below the front segment of the handle when the handle is in the first position providing clearance for the corresponding pry surface in the image forming device during removal of the toner container from the corresponding toner container receptacle in the image forming device. In some embodiments, the notch in the front of the body includes a tapered bottom surface that tapers upward in a direction from the front of the body toward the rear of the body. In some embodiments, the notch in the front of the body includes first and second tapered side surfaces that taper inward toward each other in the direction from the front of the body toward the rear of the body. In some embodiments, the notch in the front of the body includes a first notch positioned proximate to the first side of the body and a second notch positioned proximate to the second side of the body, and a vertical position of the first notch may be aligned with a vertical position of the second notch.

Some embodiments include a notch in the rear of the body providing clearance for a user's hand permitting a user to manually actuate the rear segment to move the handle from the first position to the second position.

A toner container for use in an image forming device according to another example embodiment includes a body having a top, a bottom, a front and a rear positioned between a first side and a second side of the body. The body has a

reservoir for holding toner. A downward facing outlet port is positioned on the front of the body for exiting toner from the toner container. A handle is pivotable relative to the body about a pivot axis between a first position and a second position. The handle includes a front segment extending along the front of the body when the handle is in the first position for contacting a corresponding pry surface in the image forming device when the handle moves from the first position to the second position for lifting the toner container from a corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device. The handle includes a rear segment extending along the rear of the body when the handle is in the first position permitting a user to lift the rear segment upward to move the handle from the first position to the second position causing the front segment to move downward to exert a downward force on the corresponding pry surface for removing the toner container from the corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device. The handle is positioned no higher than the top of the housing when the handle is in the first position.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings incorporated in and forming a part of the specification, illustrate several aspects of the present disclosure, and together with the description serve to explain the principles of the present disclosure.

FIG. **1** is a schematic view of an image forming device according to one example embodiment.

FIG. **2** is a perspective view of four imaging stations each having a toner cartridge and a developer unit for use with the image forming device according to one example embodiment.

FIGS. **3** and **4** are perspective views of a first toner cartridge according to one example embodiment.

FIG. **5**A is a side elevation view of the toner cartridge shown in FIGS. **3** and **4** showing a handle of the toner cartridge in a first position.

FIG. **5**B is a side elevation view of the toner cartridge shown in FIGS. **3** and **4** showing the handle of the toner cartridge in a second position.

FIGS. **6** and **7** are perspective views of a second toner cartridge according to one example embodiment.

FIG. **8**A is a side elevation view of the toner cartridge shown in FIGS. **6** and **7** showing a handle of the toner cartridge in a first position.

FIG. **8**B is a side elevation view of the toner cartridge shown in FIGS. **6** and **7** showing the handle of the toner cartridge in a second position.

FIG. **9** is a perspective view of the image forming device with an access door of the image forming device omitted showing four toner cartridges positioned within the image forming device, with three toner cartridges in installed positions and one toner cartridge in a raised position, according to one example embodiment.

FIG. **10** is a perspective view of the image forming device shown in FIG. **9** with the access door of the image forming device omitted and the four toner cartridges removed from the image forming device according to one example embodiment.

FIGS. **11**A-**11**F are sequential side cross-sectional views showing the removal of a toner cartridge from the image forming device according to one example embodiment.

FIG. **12** is a perspective cross-sectional view showing the alignment between a pry surface of the image forming device with a cutout of the toner cartridge in the position shown in FIG. **11**A according to one example embodiment.

FIG. **13** is a perspective cross-sectional view showing the alignment between the pry surface of the image forming device with the cutout of the toner cartridge in the position shown in FIG. **11**E according to one example embodiment.

DETAILED DESCRIPTION

In the following description, reference is made to the accompanying drawings where like numerals represent like elements. The embodiments are described in sufficient detail to enable those skilled in the art to practice the present disclosure. It is to be understood that other embodiments may be utilized and that process, electrical, and mechanical changes, etc., may be made without departing from the scope of the present disclosure. Examples merely typify possible variations. Portions and features of some embodiments may be included in or substituted for those of others. The following description, therefore, is not to be taken in a limiting sense and the scope of the present disclosure is defined only by the appended claims and their equivalents.

FIG. **1** illustrates a schematic view of the interior of an example image forming device **20**. Image forming device **20** includes a housing **22**. Housing **22** includes one or more input trays **28** positioned therein. Each tray **28** is sized to contain a stack of media sheets. As used herein, the term media is meant to encompass not only paper but also labels, envelopes, fabrics, photographic paper or any other desired substrate. Trays **28** are preferably removable for refilling. A control panel **30** may be located on housing **22**. Using control panel **30**, a user is able to enter commands and generally control the operation of image forming device **20**. For example, a user may enter commands to switch modes (e.g., color mode, monochrome mode), view the number of images printed, etc. A media path **32** extends through image forming device **20** for moving the media sheets through the image transfer process. Media path **32** includes a simplex path **34** and may include a duplex path **36**. A media sheet is introduced into simplex path **34** from tray **28** by a pick mechanism **38**. In the example embodiment shown, pick mechanism **38** includes a roll **40** positioned at the end of a pivotable arm **42**. Roll **40** rotates to move the media sheet from tray **28** and into media path **32**. The media sheet is then moved along media path **32** by various transport rollers. Media sheets may also be introduced into media path **32** by a manual feed **46** having one or more rolls **48**.

Image forming device **20** includes an image transfer section that includes one or more imaging stations **50**. Each imaging station **50** includes a toner cartridge **100**, a developer unit **200** and a photoconductive unit (PC unit) **300**. Each toner cartridge **100** includes a reservoir **102** for holding toner and an outlet port in communication with an inlet port of a corresponding developer unit **200** for transferring toner from reservoir **102** to developer unit **200**. In the example embodiment illustrated, developer unit **200** utilizes what is commonly referred to as a single component development system. In this embodiment, each developer unit **200** includes a toner reservoir **202** and a toner adder roll **204** that moves toner from reservoir **202** to a developer roll **206**. In another embodiment, developer unit **200** utilizes what is commonly referred to as a dual component development system. In this embodiment, toner in toner reservoir **202** is mixed with magnetic carrier beads. The magnetic carrier beads may be coated with a polymeric film to provide triboelectric properties to attract toner to the carrier beads as the toner and the magnetic carrier beads are mixed in the toner reservoir. In this embodiment, developer roll **206** attracts the magnetic carrier beads having toner thereon to developer roll **206** through the use of magnetic fields. Each PC unit **300** includes a charging roll **304** and a photoconductive (PC) drum **302** for each imaging station **50**. PC drums **302** are mounted substantially parallel to each other. For purposes of clarity, developer unit **200** and PC unit **300** are labeled on only one of the imaging stations **50**. In the example embodiment illustrated, each imaging station **50** is substantially the same except for the color or type of toner.

Each charging roll **304** forms a nip with the corresponding PC drum **302**. During a print operation, charging roll **304** charges the surface of PC drum **302** to a specified voltage. A laser beam from a printhead **52** associated with each imaging station **50** is then directed to the surface of PC drum **302** and selectively discharges those areas it contacts to form a latent image. Developer roll **206** then transfers toner to PC drum **302** to form a toner image. A metering device, such as a doctor blade, may be used to meter toner on developer roll **206** and apply a desired charge to the toner prior to its transfer to PC drum **302**. Toner is attracted to the areas of PC drum **302** surface discharged by the laser beam from printhead **52**.

In the example embodiment illustrated, an intermediate transfer mechanism (ITM) **54** is disposed adjacent to imaging stations **50**. In this embodiment, ITM **54** is formed as an endless belt trained about a drive roll **56**, a tension roll **58** and a back-up roll **60**. During print operations, ITM **54** moves past imaging stations **50** in a clockwise direction as viewed in FIG. **1**. One or more of PC drums **302** apply toner images in their respective colors to ITM **54** at a first transfer nip **62**. ITM **54** rotates and collects the one or more toner images from imaging stations **50** and then conveys the toner images to a media sheet advancing through simplex path **34** at a second transfer nip **64** formed between a transfer roll **66** and ITM **54**, which is supported by back-up roll **60**. In other embodiments, the toner image is transferred to the media sheet directly by the PC drum(s) **302**.

The media sheet with the toner image is then moved along simplex path **34** and into a fuser area **68**. Fuser area **68** includes fusing rolls or belts **70** that form a nip **72** to adhere the toner image to the media sheet. The fused media sheet then passes through exit rolls **74** located downstream from fuser area **68**. Exit rolls **74** may be rotated in either forward or reverse directions. In a forward direction, exit rolls **74** move the media sheet from simplex path **34** to an output area **76** of image forming device **20**. In a reverse direction, exit rolls **74** move the media sheet into duplex path **36** for image formation on a second side of the media sheet.

A monocolor image forming device **20** may include a single imaging station **50**, as compared to a color image forming device **20** that may include multiple imaging stations **50**. FIG. **2** illustrates a set of four imaging stations **50** that each includes a respective toner cartridge **100**, developer unit **200** and PC unit **300**. In the example embodiment illustrated, image forming device **20** includes three toner cartridges **100***a*, **100***b*, **100***c* having substantially the same construction and size and a toner cartridge **100***d* having a larger overall size and toner capacity. In this embodiment, toner cartridges **100***a*, **100***b*, **100***c* each contain colored toner, such as, for example, cyan, yellow and magenta toner, respectively, and toner cartridge **100***d* contains black toner. In some embodiments, image forming device **20** may also permit installation of a toner cartridge having substantially the same construction and size as toner cartridges **100***a*, **100***b*, **100***c* in place of the larger toner cartridge **100***d* such that image forming device **20** may permit the use of either a smaller (normal capacity) black toner cartridge or a larger capacity black toner cartridge, depending on the user's selection.

FIGS. **3** and **4** show toner cartridge **100***a*, which may have substantially the same construction as toner cartridges **100***b* and **100***c* and, optionally, toner cartridge **100***d*, according to one example embodiment. Toner cartridge **100***a* includes a body **104** housing toner reservoir **102** therein. Body **104** includes a top **105**, a bottom **106**, a front **107**, a rear **108** and a pair of sides **109**, **110**. In the example embodiment illustrated, body **104** includes a main section **112** and an extension section **114**. Extension section **114** is positioned at the bottom **106** of body **104**. As illustrated in FIGS. **3** and **4**, in one example embodiment, a depth of extension section **114** is less than a depth of main section **112**, and a height of extension section **114** is less than a height of main section **112**.

Toner cartridge **100***a* includes an outlet port **116** for transferring toner to an inlet port of the corresponding developer unit **200**. In the example embodiment illustrated, outlet port **116** is formed as a downward facing opening on main section **112** on front **107** of body **104**, roughly midway up the front **107** of body **104**. Outlet port **116** may include a shutter that regulates whether toner is permitted to flow from toner reservoir **102** out of outlet port **116**.

Toner cartridge **100***a* further includes a drive gear **118** that meshes with and receives rotational power from a corresponding gear in image forming device **20**, e.g., on the corresponding developer unit **200**, when toner cartridge **100***a* is installed in image forming device **20** in order to provide rotational power to various toner agitators positioned within reservoir **102** for moving toner to outlet port **116**. In the example embodiment illustrated, drive gear **118** is positioned on the front **107** of body **104**, and a bottom portion of drive gear **118** mates with the corresponding gear in image forming device **20** when toner cartridge **100***a* is installed in image forming device **20**.

Toner cartridge **100***a* may also include various alignment members **120** that align toner cartridge **100***a* with the corresponding developer unit **200** during insertion of toner cartridge **100***a* in the direction shown by arrow A in FIG. **3**. For example, alignment members **120** may include a combination of projections that project outward from sides **109**, **110** of body **104** and/or elongated slots formed as depressions in sides **109**, **110** that mate with corresponding slots and/or projections, respectively, to ensure accurate positioning of toner cartridge **100***a*. For example, alignment members **120** help ensure that outlet port **116** mates with the inlet port of developer unit **200** and that drive gear **118** mates with the corresponding drive gear in image forming device **20**.

Toner cartridge **100***a* may also include an electrical connector **122** having electrical contacts that are matable with corresponding electrical contacts in image forming device **20** when toner cartridge **100***a* is installed in image forming device **20** for facilitating a communication link between processing circuitry of toner cartridge **100***a* and a controller of image forming device **20**. The processing circuitry of toner cartridge **100***a* may provide authentication functions, safety and operational interlocks, operating parameters and/or usage information related to toner cartridge **100***a* and may include hardware and/or software logic as desired. Electrical connector **122**, including the electrical contacts, may be positioned in a recess or cavity formed in the bottom **106** of body **104**. An opening at a bottom end of the cavity and a bottommost portion of body **104** permits a corresponding electrical connector in image forming device **20** to enter the cavity during downward insertion of toner cartridge **100***a* into image forming device **20**.

With reference to FIGS. **3**, **4**, **5**A and **5**B, toner cartridge **100***a* includes a handle **130** for assisting a user with removing toner cartridge **100***a* from image forming device **20**. Handle **130** is pivotable relative to body **104** about a pivot axis **132** between a first position shown in FIGS. **3**, **4** and **5**A and a second position shown in FIG. **5**B. In the example embodiment illustrated, pivot axis **132** is horizontal and extends from side **109** to side **110**, generally parallel to front **107** and rear **108** of body **104**. Pivot axis **132** is positioned near top **105** of body **104** (for example, spaced slightly below top **105** of body **104**). In the embodiment illustrated, pivot axis **132** is positioned closer to front **107** of body **104** than to rear **108** of body **104**. In the example embodiment illustrated, handle **130** tends to fall due to gravity from the second position shown in FIG. **5**B and settle at the first position shown in FIG. **5**A. In other embodiments, handle **130** may be biased, for example, by a spring or other resilient biasing member, toward the first position, as desired.

Handle **130** includes a front segment **134** that extends along front **107** of body **104**. In the example embodiment illustrated, front segment **134** extends horizontally along front **107** of body **104**. Front segment **134** includes a bottom contact surface **136** that contacts a corresponding pry surface in image forming device **20** when handle **130** is manually moved by a user from the first position to the second position in order to remove toner cartridge **100***a* from image forming device **20** as discussed in greater detail below. In the example embodiment illustrated, front segment **134** including bottom contact surface **136** is positioned higher than pivot axis **132** when handle **130** is in the first position shown in FIG. **5**A and lower than pivot axis **132** when handle **130** is in the second position shown in FIG. **5**B. In this embodiment, a top surface **137** of front segment **134** is positioned no higher than top **105** of body **104** in the first and second positions of handle **130**.

Handle **130** also includes a rear segment **138** that extends along rear **108** of body **104**. In the example embodiment illustrated, rear segment **138** extends horizontally along rear **108** of body **104**. Rear segment **138** provides a contact point for a user's hand or finger to allow the user to actuate handle **130** and move handle **130** from the first position to the second position in order to remove toner cartridge **100***a* from image forming device **20** as discussed in greater detail below. In the example embodiment illustrated, rear segment **138** is positioned lower than pivot axis **132** when handle **130** is in the first position shown in FIG. **5**A and higher than pivot axis **132** when handle **130** is in the second position shown in FIG. **5**B. In this embodiment, rear segment **138** is positioned lower than top **105** of body **104** in the first position of handle **130**, and rear segment **138** extends higher than top **105** of body **104** in the second position of handle **130**.

In the example embodiment illustrated, handle **130** includes a pair of side segments **140**, **142** that extend along respective sides **109**, **110** of body **104** and connect front segment **134** to rear segment **138**. In the example embodiment illustrated, each side segment **140**, **142** includes a respective front portion **140***a*, **142***a* proximate to front **107** of body **104** that extends downward and rearward along the respective sides **109**, **110** of body **104** when handle **130** is in the first position and a respective rear portion **140***b*, **142***b* proximate to rear **108** of body **104** that extends horizontally along the respective sides **109**, **110** of body **104** when handle **130** is in the first position. In the example embodiment illustrated, pivot axis **132** extends through front portions **140***a*, **142***a* of side segments **140**, **142**.

Handle **130** may be pivotably mounted to body **104** by any suitable means. For example, in one embodiment, a post on an inner surface of each side segment **140**, **142** of handle **130** is received by a corresponding mount on each side **109**, **110** of body **104** with the engagement between the posts and corresponding mounts defining pivot axis **132** of handle **130**.

With reference to FIGS. **5**A and **5**B, in order to actuate handle **130** and move handle **130** from the first position shown in FIG. **5**A to the second position shown in FIG. **5**B, a user typically contacts rear segment **138** of handle **130** (for example, a bottom surface of rear segment **138**) and lifts rear segment **138** upward. When handle **130** moves from the first position to the second position, handle **130** pivots clockwise as viewed in FIGS. **5**A and **5**B about pivot axis **132**. When handle **130** moves from the first position to the second position, rear segment **138** raises from a position below pivot axis **132** and top **105** of body **104** to a position above pivot axis **132** and top **105** of body **104**, and front segment **134** lowers from a position above pivot axis **132** to a position below pivot axis **132**. When handle **130** returns from the second position to the first position, for example, as a result of gravity when a user releases handle **130**, handle **130** pivots counterclockwise as viewed in FIGS. **5**A and **5**B about pivot axis **132**.

With reference back to FIG. **3**, in the example embodiment illustrated, toner cartridge **100***a* includes a notch or cutout **150** in front **107** of body **104** that provides clearance for a corresponding pry surface in image forming device **20** in order to accommodate the pry surface during removal of toner cartridge **100***a* from image forming device **20** as discussed in greater detail below. Cutout **150** is positioned below contact surface **136** of front segment **134** of handle **130** when handle **130** is in the first position. In the example embodiment illustrated, cutout **150** is formed by tapered side surfaces **152**, **153**, a tapered bottom surface **154** and an open top end **156**. Side surfaces **152**, **153** taper inward toward each other in a direction from front **107** to rear **108**, and bottom surface **154** tapers upward in a direction from front **107** to rear **108**.

With reference to FIG. **4**, in the example embodiment illustrated, toner cartridge **100***a* includes a notch or cutout **160** in rear **108** of body **104** that provides clearance for a user's hand. As discussed in greater detail below, cutout **160** makes it easier for a user to engage or grasp rear segment **138** of handle **130** when handle **130** is in the first position in order to actuate handle **130** and move handle **130** from the first position to the second position. In the example embodiment illustrated, cutout **160** includes an inset, tapered rear surface **162** that extends further toward front **107** of body **104** as surface **162** extends downward such that a depth of cutout **160** along a direction from rear **108** to front **107** increases in a downward direction.

FIGS. **6**, **7**, **8**A and **8**B show toner cartridge **100***d*, which has a larger overall size and toner capacity than toner cartridge **100***a* discussed above, according to one example embodiment. For the sake of brevity, the description of features common to both toner cartridge **100***a* and toner cartridge **100***d* is not repeated, and like reference numbers are used to designate like elements.

Toner cartridge **100***d* includes a body **1104** that is substantially the same as body **104** of toner cartridge **100***a*, except that body **1104** has a larger overall size and toner capacity than body **104**. Like body **104**, body **1104** includes a top **1105**, a bottom **1106**, a front **1107**, a rear **1108** and a pair of sides **1109**, **1110**. Toner cartridge **100***d* includes a handle **1130** for aiding the removal of toner cartridge **100***d* from image forming device **20**. Handle **1130** is pivotable relative to body **1104** about pivot axis **132** between a first position shown in FIGS. **6**, **7** and **8**A and a second position shown in FIG. **8**B. In the example embodiment illustrated, handle **1130** of toner cartridge **100***d* is substantially the same as handle **130** of toner cartridge **100***a*, except that a front segment **1134** and a rear segment **1138** are wider (in a direction from side **1109** to side **1110**) than front segment **134** and rear segment **138** of handle **130** in order to accommodate the larger overall size of body **1104**.

In the example embodiment illustrated, handle **1130** is operated in the same manner as handle **130**. In order to move handle **1130** from the first position shown in FIG. **8**A to the second position shown in FIG. **8**B, a user typically contacts rear segment **1138** of handle **1130** and lifts rear segment **1138** upward. When handle **1130** moves from the first position to the second position, handle **1130** pivots clockwise as viewed in FIGS. **8**A and **8**B about pivot axis **132**. When handle **1130** returns from the second position to the first position, for example, as a result of gravity when a user releases handle **1130**, handle **1130** pivots counterclockwise as viewed in FIGS. **8**A and **8**B about pivot axis **132**.

With reference back to FIG. **6**, in the example embodiment illustrated, toner cartridge **100***d* includes a pair of notches or cutouts **1150***a*, **1150***b* in front **1107** of body **1104** that provide clearance for corresponding pry surfaces in image forming device **20** in order to accommodate the pry surfaces during removal of toner cartridge **100***d* from image forming device **20**. In the example embodiment illustrated, each cutout **1150***a*, **115***b* is substantially the same as cutout **150** discussed above. In this embodiment, a vertical position of cutout **1150***a* is aligned with a vertical position of cutout **1150***b*. Cutout **1150***a* is positioned proximate to side **1109** of body **1104**, and cutout **1150***b* is positioned proximate to side **1110** of body **1104**.

With reference to FIG. **7**, in the example embodiment illustrated, toner cartridge **100***d* includes a notch or cutout **1160** in rear **1108** of body **1104** that provides clearance for a user's hand. In the example embodiment illustrated, cutout **1160** is substantially the same as cutout **160** discussed above.

FIGS. **9** and **10** show image forming device **20** according to one example embodiment. In this embodiment, image forming device **20** includes a multi-function device including an electrophotographic printer **400** and a document scanner **402**. FIG. **9** shows image forming device **20** with an access door omitted in order to show toner cartridges **100***a*, **100***b*, **100***c*, **100***d* installed in corresponding toner cartridge receptacles **404***a*, **404***b*, **404***c*, **404***d* of image forming device **20**. When the access door of image forming device **20** is opened, toner cartridges **100***a*, **100***b*, **100***c*, **100***d* are accessible within a recessed area **408** of image forming device **20**. In FIG. **9**, toner cartridges **100***b*, **100***c*, **100***d* are fully installed and seated in toner cartridge receptacles **404***b*. **404***c*, **404***d* of image forming device **20** with handles **130**, **1130** of toner cartridges **100***b*, **100***c*, **100***d* in their respective first positions. In FIG. **9**, handle **130** of toner cartridge **100***a* is shown moved from the first position to the second position causing bottom contact surface **136** of front segment **134** to contact a corresponding pry surface **406***a* positioned within image forming device **20** adjacent to toner cartridge receptable **404***a*, in turn causing toner cartridge **100***a* to raise vertically from toner cartridge receptacle **404***a* as discussed in greater detail below.

FIG. **10** shows image forming device **20** with the access door omitted and toner cartridges **100***a*, **100***b*, **100***c*, **100***d* removed. Toner cartridge receptacles **404***a*, **404***b*, **404***c*, **404***d* are shown positioned to receive and retain corresponding toner cartridges **100***a*, **100***b*, **100***c*, **100***d* within image forming device **20**. In the example embodiment illustrated, toner cartridge receptacle **404***d* is positioned and sized to receive a relatively large toner cartridge **100***d*, such as the embodiment shown in FIGS. **6-8**B, or a smaller toner cartridge **100***d*, such as the embodiment shown in FIGS. **3-5**B. Pry surfaces **406***a*, **406***b*, **406***c*, **406***d*, **406***e* are shown positioned to engage bottom contact surfaces **136**, **1136** of front segments **134**, **1134** of handles **130**, **1130** of toner cartridges **100***a*, **100***b*, **100***c*, **100***d* to permit a user to remove toner cartridges **100***a*, **100***b*, **100***c*, **100***d* from image forming device **20**. In the example embodiment illustrated, toner cartridge receptacle **404***d* includes a pair of pry surfaces **406***d*, **406***e* positioned adjacent thereto. If a relatively large toner cartridge **100***d*, such as the embodiment shown in FIGS. **6-8**B, is installed in toner cartridge receptacle **404***d*, both pry surfaces **406***d*, **406***e* contact bottom contact surface **1136** on front segment **1134** of handle **1130** when handle **1130** is moved from the first position to the second position in order to prevent twisting or flexing of handle **1130** during removal of toner cartridge **100***d*. If a smaller toner cartridge **100***d*, such as the embodiment shown in FIGS. **3-5**B, is installed in toner cartridge receptacle **404***d*, only pry surface **406***d* contacts bottom contact surface **136** on front segment **134** of handle **130** when handle **130** is moved from the first position to the second position.

FIGS. **11**A-**11**F are sequential views showing the removal of toner cartridge **100***a* from image forming device **20**, including the operation of handle **130**, according to one example embodiment. The removal of toner cartridges **100***b*, **100***c*, **100***d* from image forming device **20**, including the operation of handle **1130** of toner cartridge **100***d*, is substantially the same. FIG. **11**A shows toner cartridge **100***a* fully installed and seated in corresponding toner cartridge receptacle **404***a* with handle **130** in the first position. When handle **130** is in the first position with toner cartridge **100***a* fully seated in toner cartridge receptacle **404***a*, bottom contact surface **136** of front segment **134** of handle **130** is shown spaced slightly above pry surface **406***a*.

FIG. **12** is a perspective view showing toner cartridge **100***a* in the position illustrated in FIG. **11**A. As shown in FIG. **12**, when handle **130** is in the first position with toner cartridge **100***a* fully seated in toner cartridge receptacle **404***a*, pry surface **406***a* in image forming device **20** is aligned along a vertical direction with cutout **150** on front **107** of body **104** of toner cartridge **100***a*, spaced above open top end **156** of cutout **150**.

With reference back to FIGS. **11**A-**11**F, as shown in FIGS. **11**B and **11**C, when a user manually lifts rear segment **138** of handle **130** upward, handle **130** pivots (counterclockwise as viewed in FIGS. **11**A-**11**F) about pivot axis **132** from the first position shown in FIG. **11**A. The rotation of handle **130** about pivot axis **132** resulting from the user lifting rear segment **138** causes handle **130** to act as a lever with bottom contact surface **136** of front segment **134** contacting and exerting a downward force on corresponding pry surface **406***a* in image forming device **20**. The upward force applied by the user to rear segment **138** of handle **130** and the upward reaction force from pry surface **406***a* on front segment **134** of handle **130** cause toner cartridge **100***a* to raise upward from its toner cartridge receptacle **404***a* as shown in FIGS. **11**B and **11**C.

As the user continues to lift rear segment **138** of handle **130** upward, handle **130** pivots further (counterclockwise as viewed in FIGS. **11**A-**11**F) about pivot axis **132** away from the first position of handle **130** and toward the second position of handle **130** as shown in FIGS. **11**D and **11**E. The contact between bottom contact surface **136** of front segment **134** of handle **130** and pry surface **406***a* causes toner cartridge **100***a* to continue to raise upward from toner cartridge receptacle **404***a* as shown in FIGS. **11**D and **11**E. FIG. **11**E shows handle **130** in the second position with bottom contact surface **136** of front segment **134** of handle **130** in contact with pry surface **406***a*.

FIG. **13** is a perspective view showing toner cartridge **100***a* in the position illustrated in FIG. **11**E. As shown in FIG. **13**, when handle **130** is in the second position with toner cartridge **100***a* raised upward from toner cartridge receptacle **404***a* and bottom contact surface **136** of front segment **134** of handle **130** is in contact with pry surface **406***a*, pry surface **406***a* extends into cutout **150** on front **107** of body **104** of toner cartridge **100***a*. In this manner, cutout **150** accommodates pry surface **406***a* and permits the use of a handle **130** that does not need to extend forward beyond the front **107** of body **104** of toner cartridge **100***a*.

With reference back to FIGS. **11**A-**11**F, after handle **130** reaches the second position, toner cartridge **100***a* tends to rest on a top portion of toner cartridge receptacle **404***a* permitting a user to manually lift and remove toner cartridge **100***a* with little resistance from toner cartridge receptacle **404***a*. FIG. **11**F shows toner cartridge **100***a* lifted and separated from toner cartridge receptacle **404***a*.

As discussed above, in the embodiment illustrated, handle **130** of toner cartridge **100***a* tends to rest at the first position due to gravity. Accordingly, during insertion of toner cartridge **100***a* into toner cartridge receptacle **404***a* of image forming device **20**, front segment **134** of handle **130** tends to be spaced above pry surface **406***a* until toner cartridge **100***a* is fully seated or nearly fully seated in toner cartridge receptacle **404***a*. If handle **130** of toner cartridge **100***a* is pivoted away from the first position and toward the second position during insertion of toner cartridge **100***a* into toner cartridge receptacle **404***a*, pry surface **406***a* will contact bottom contact surface **136** of front segment **134** of handle **130**, and the downward insertion of toner cartridge **100***a* into toner cartridge receptacle **404***a* will cause handle **130** to pivot about pivot axis **132** toward the first position. When toner cartridge **100***a* is fully installed and seated in toner cartridge receptacle **404***a*, front segment **134** of handle **130** tends to rest on or just above pry surface **406***a* with handle **130** in the first position as illustrated in FIGS. **11**A and **12**.

Accordingly, the present disclosure provides an assembly for efficiently removing an existing toner container and installing a replacement toner container. Handles **130**, **1130** of the present disclosure allow a reduced overall height of toner cartridge **100** in comparison with a handle positioned on the top of toner cartridge **100**. For example, in the embodiments illustrated, handles **130**, **1130**, including front segments **134**, **1134**, rear segments **138**, **1138** and side segments **140**, **142**, are positioned no higher than top **105** of body **104** in the first position of handle **130**. As shown in FIG. **9**, where image forming device **20** includes one or more components positioned above toner cartridges **100***a*, **100***b*, **100***c*, **100***d*, such as document scanner **402**, handles **130**, **1130** of the present disclosure allow the overall height of image forming device **20** to be reduced, consistent with user preferences for smaller devices. If, on the other hand, toner cartridges **100***a*, **100***b*, **100***c*, **100***d* included handles occupying more vertical space above toner cartridges **100***a*, **100***b*, **100***c*, **100***d*, such as handles positioned on the tops of toner cartridges **100***a*, **100***b*. **100***c*, **100***d*, a larger recessed area **408** having a greater height would be required in order to provide users with enough vertical space to remove and install toner cartridges **100***a*, **100***b*, **100***c*, **100***d*.

As illustrated in FIG. **9**, handles **130**, **1130** of the present disclosure also provide the user with a touch point, i.e., rear segment **138**, **1138** in the example embodiments illustrated, that is positioned closer to the user than a handle positioned on the top of the toner cartridge. In this manner, handles **130**, **1130** of the present disclosure provide a more visible and more intuitive mechanism for removing each toner cartridge **100**.

Further, handles **130**, **1130** of the present disclosure provide a relatively long lever arm for removing toner cartridge **100** that extends from rear segment **138**, **1138** actuated by the user to front segment **134**, **1134** that contacts the corresponding pry surface in image forming device **20**. The relatively long lever arm provided by handles **130**, **1130** helps reduce the force required for a user to move handle **130**, **1130** from the first position to the second position to lift toner cartridge **100** out of its corresponding toner cartridge receptacle within image forming device **20**.

Although the example embodiment shown in FIG. **2** includes toner cartridges **100**, developer units **200** and PC units **300** positioned in separate replaceable units, it will be appreciated that the replaceable unit(s) of image forming device **20** may employ any suitable configuration as desired. For example, in one embodiment, the main toner supply for image forming device **20**, developer unit **200** and PC unit **300** are combined in a separate replaceable unit for each color toner. In another embodiment, the main toner supply for image forming device **20** and developer unit **200** are provided in a first replaceable unit and PC unit **300** is provided in a second replaceable unit. Other combinations are possible without departing from the scope of the present disclosure.

The foregoing description illustrates various aspects of the present disclosure. It is not intended to be exhaustive. Rather, it is chosen to illustrate the principles of the present disclosure and its practical application to enable one of ordinary skill in the art to utilize the present disclosure, including its various modifications that naturally follow. All modifications and variations are contemplated within the scope of the present disclosure as determined by the appended claims. Relatively apparent modifications include combining one or more features of various embodiments with features of other embodiments.

The invention claimed is:

1. A toner container for use in an image forming device, comprising:

a body having a top and a bottom relative to an operative orientation of the toner container when the toner container is installed in the image forming device, the body having a first side, a second side, a third side and a fourth side, the second side is opposite the first side, the fourth side is opposite the third side, the body has a reservoir for holding toner;

a downward facing outlet port positioned on the first side of the body for exiting toner from the toner container;

a handle pivotable relative to the body about a pivot axis between a first position and a second position, the handle includes a front segment extending along the first side of the body when the handle is in the first position for contacting a corresponding pry surface in the image forming device when the handle moves from the first position to the second position for lifting the toner container from a corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device; and a notch in the first side of the body below the front segment of the handle when the handle is in the first position providing clearance for the corresponding pry surface in the image forming device during removal of the toner container from the corresponding toner container receptacle in the image forming device.

2. The toner container of claim **1**, wherein when the toner container is in the operative orientation, the front segment extends horizontally along the first side of the body when the handle is in the first position.

3. The toner container of claim **1**, wherein when the toner container is in the operative orientation, a bottom contact surface of the front segment positioned to contact the corresponding pry surface in the image forming device when the handle moves from the first position to the second position is positioned higher than the pivot axis when the handle is in the first position and lower than the pivot axis when the handle is in the second position.

4. The toner container of claim **1**, wherein the handle includes a rear segment extending along the second side of the body when the handle is in the first position permitting a user to actuate the rear segment to move the handle from the first position to the second position for removing the toner container from the corresponding toner container receptacle in the image forming device when the toner container is installed in the image forming device.

5. The toner container of claim **4**, wherein when the toner container is in the operative orientation, the rear segment extends horizontally along the second side of the body when the handle is in the first position.

6. The toner container of claim **4**, wherein when the toner container is in the operative orientation, the rear segment is positioned lower than the pivot axis when the handle is in the first position and higher than the pivot axis when the handle is in the second position.

7. The toner container of claim **4**, further comprising a notch in the second side of the body providing clearance for a user's hand permitting a user to manually actuate the rear segment to move the handle from the first position to the second position.

8. The toner container of claim **1**, wherein the notch in the first side of the body includes a tapered bottom surface that tapers upward in a direction from the first side of the body toward the second side of the body.

9. The toner container of claim **8**, wherein the notch in the first side of the body includes first and second tapered side surfaces that taper inward toward each other in the direction from the first side of the body toward the second side of the body.

10. The toner container of claim **1**, wherein the notch in the first side of the body includes a first notch positioned proximate to the third side of the body and a second notch positioned proximate to the fourth side of the body.

11. The toner container of claim **10**, wherein when the toner container is in the operative orientation, a vertical position of the first notch is aligned with a vertical position of the second notch.

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